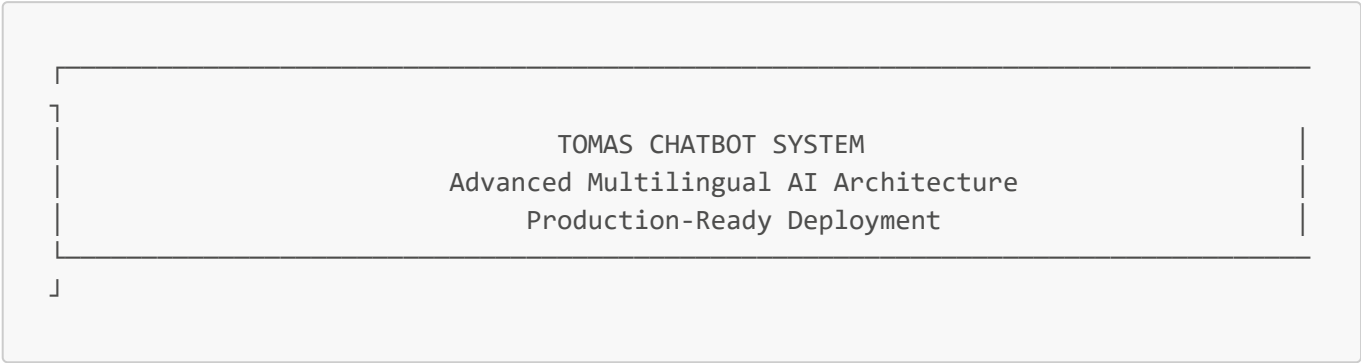
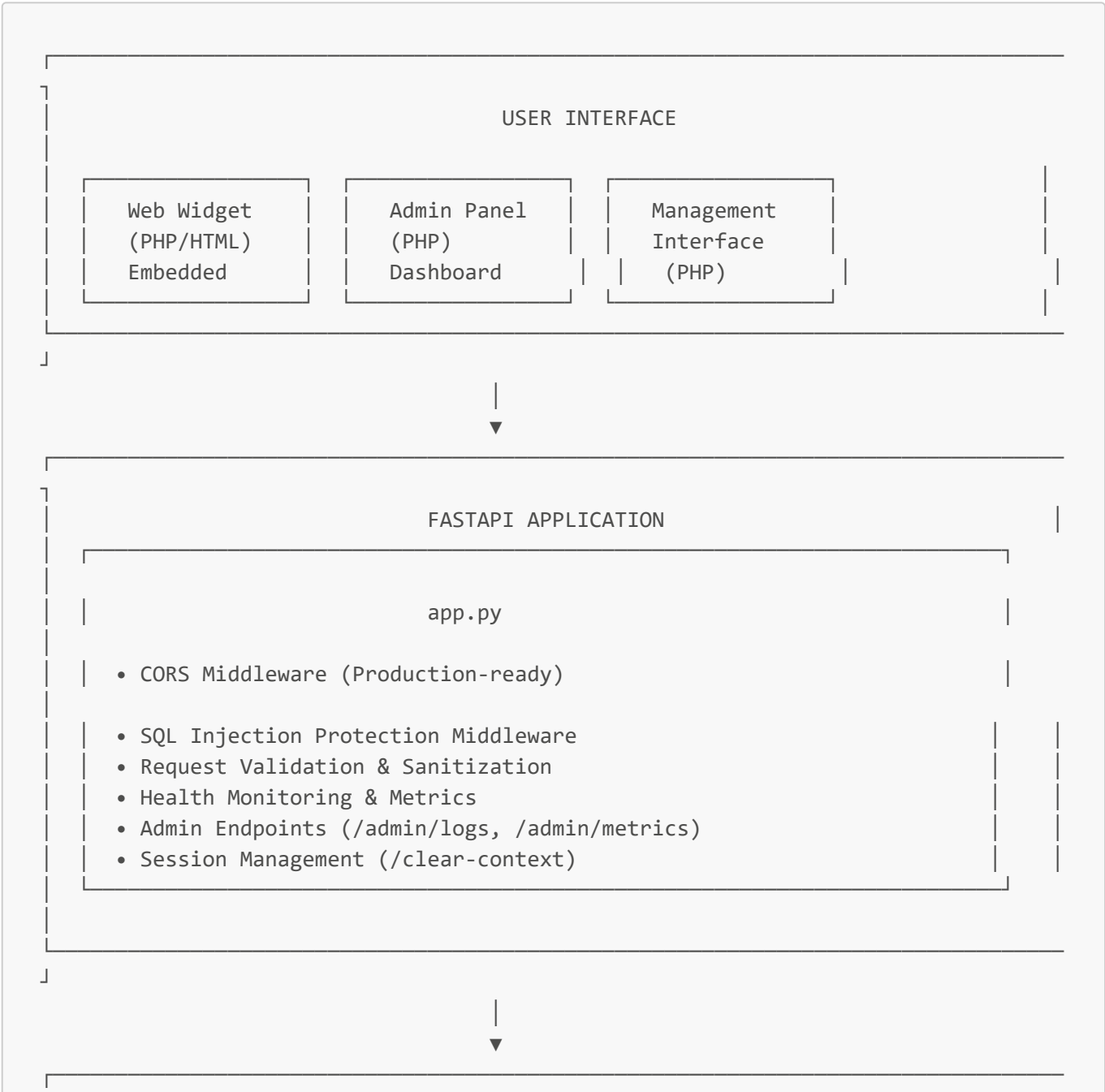


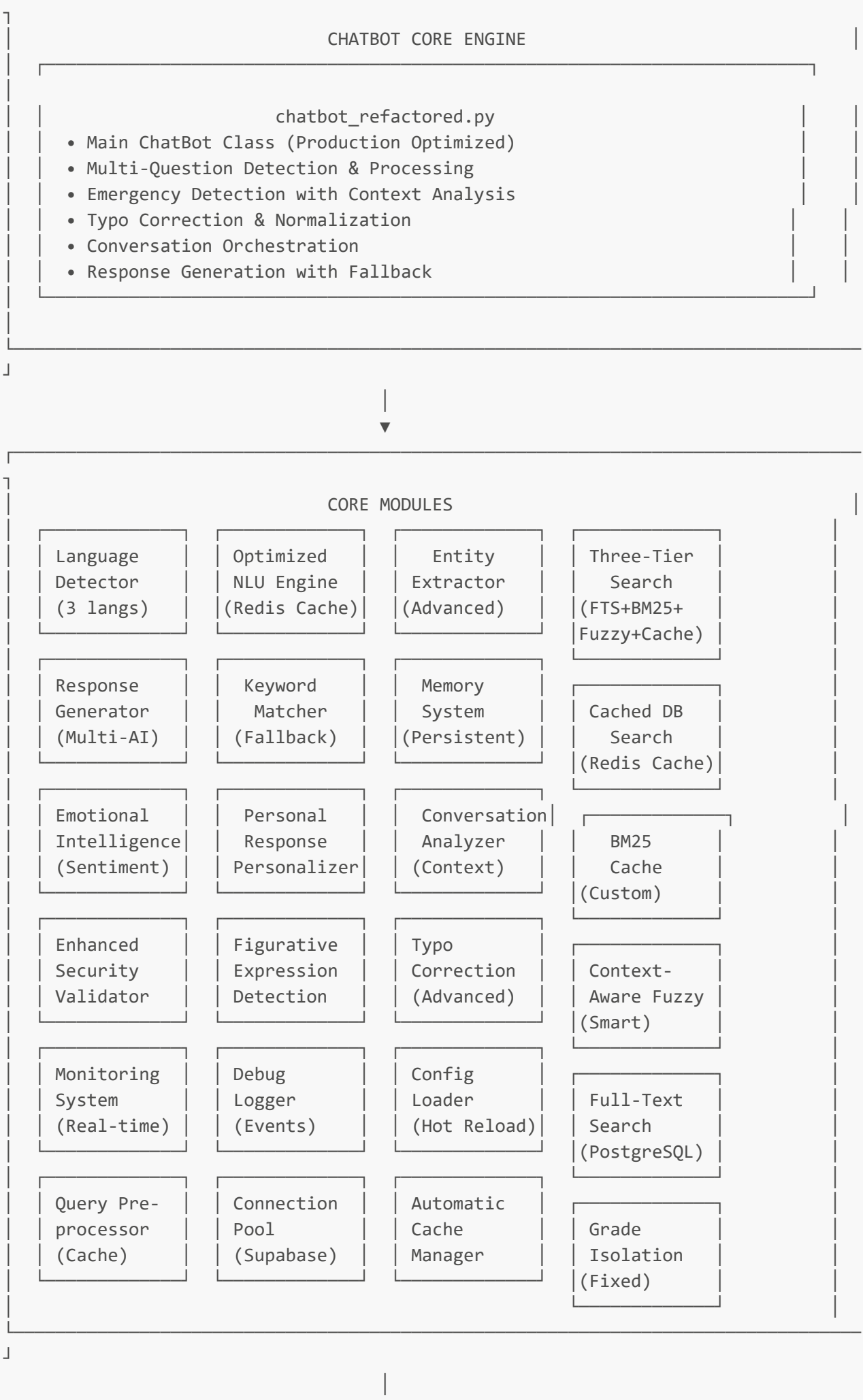
TOMAS Chatbot Architecture Diagram (Updated)

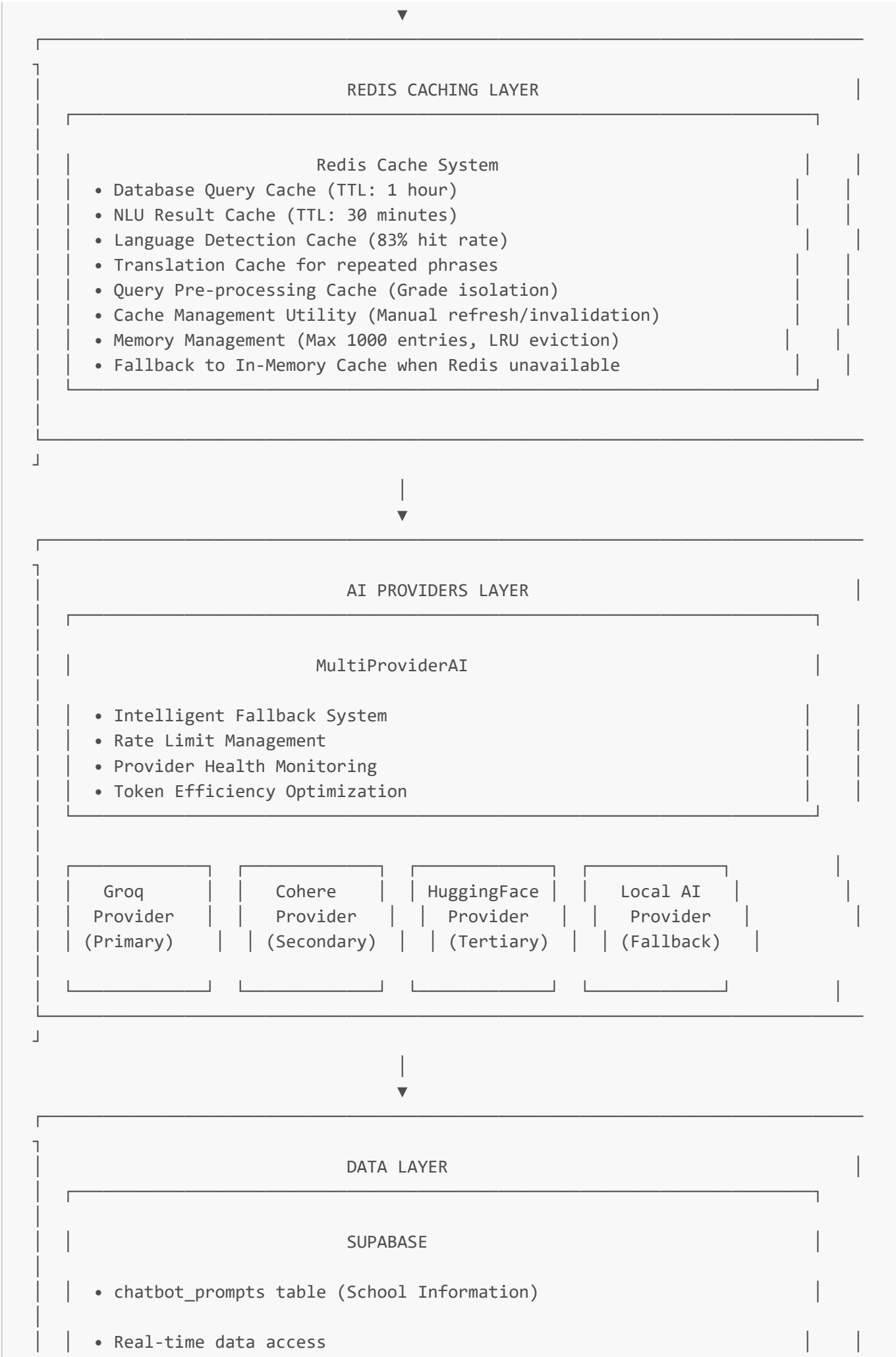
System Overview



Core Architecture







- Advanced search with ImprovedScorer
- Semantic similarity matching
- Context-aware result ranking
- Connection Pooling (Transaction Mode)
- PgBouncer optimization (3-5x efficiency)

AI Provider Configuration (Updated)

AI PROVIDER HIERARCHY

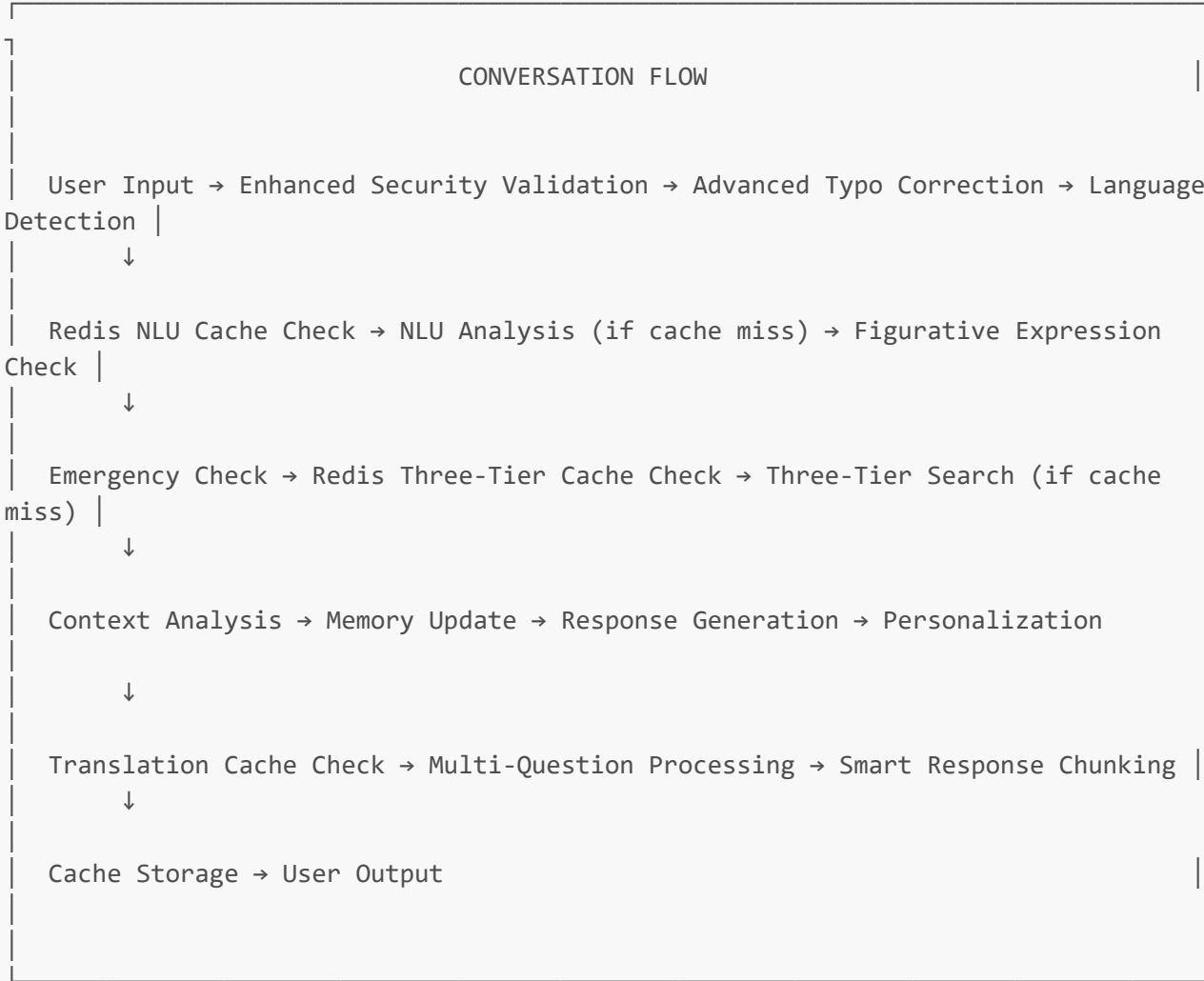
1. GROQ PROVIDER (Primary)
 - Model: meta-llama/llama-4-scout-17b-16e-instruct
 - API Key: GROQ_API_KEY
 - Rate Limits: 30 requests/minute
 - Fallback: CohereProvider
 - Token Efficiency: Optimized prompts
2. COHERE PROVIDER (Secondary)
 - Model: command-a-03-2025
 - API Key: COHERE_API_KEY
 - Rate Limits: 1000 requests/minute
 - Fallback: HuggingFaceProvider
3. HUGGING FACE PROVIDER (Tertiary)
 - Model: deepseek-ai/DeepSeek-V3-0324
 - API Key: HUGGINGFACE_API_KEY (optional)
 - | Rate Limits: 1000 requests/day
 - Fallback: LocalAIProvider
4. LOCAL AI PROVIDER (Final Fallback)
 - Model: deepseek-ai/DeepSeek-V3-0324
 - API Key: HUGGINGFACE_API_KEY (optional)
 - Rate Limits: Unlimited (free)
 - Fallback: Context-aware keyword responses

Core Features & Capabilities (Updated)

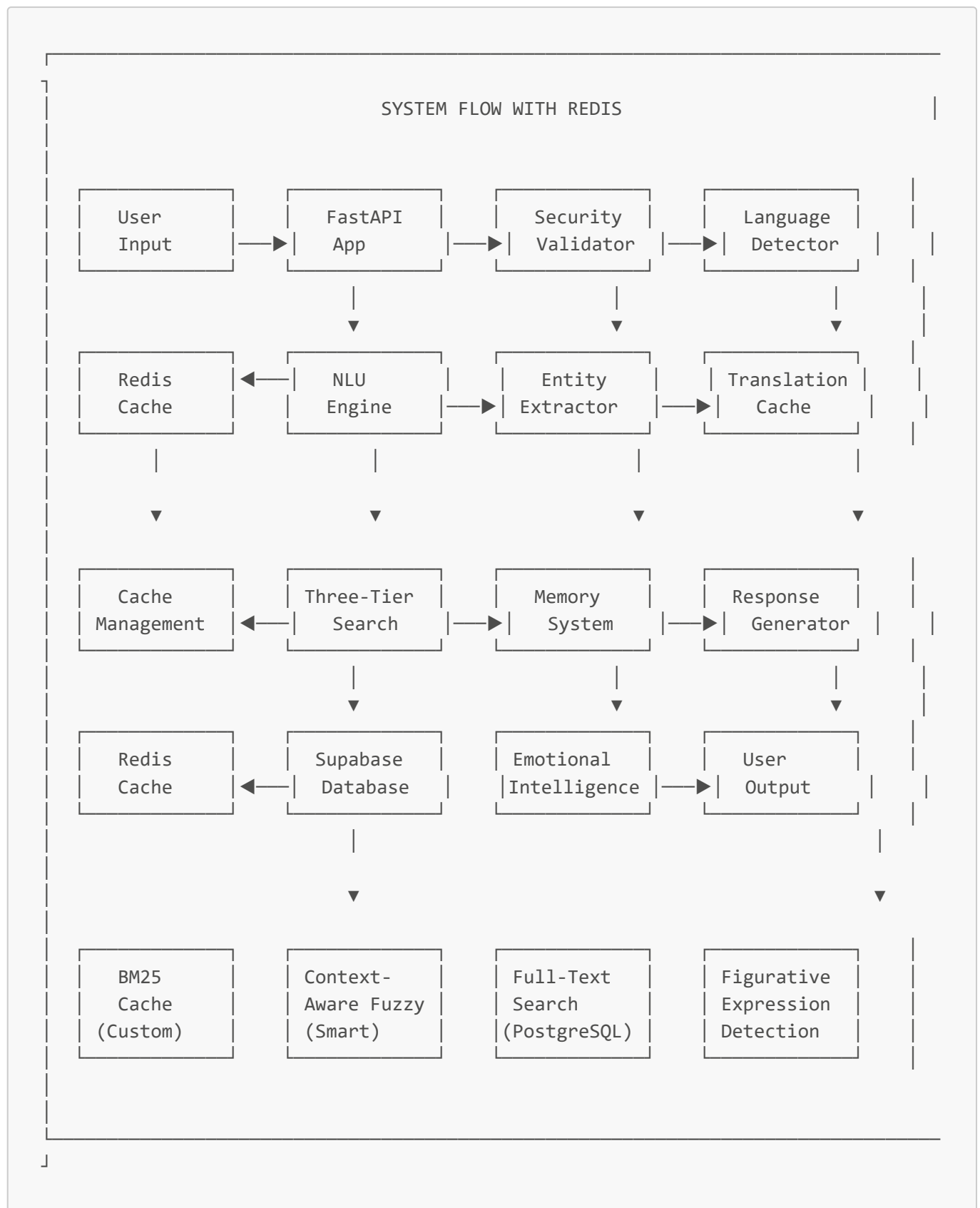
CHATBOT FEATURES	
 MULTILINGUAL SUPPORT - English (Primary) - Tagalog (Secondary) - Aklanon (Tertiary) - Mixed-language detection with confidence scoring - Context-aware translation with grammar preservation - Language-specific response formatting	
 INTELLIGENT CONVERSATION - Natural Language Understanding (NLU) with 40+ intents - Advanced Entity Extraction (Names, Grades, Subjects, Relationships) - Conversation Memory with persistent user profiles - Context Awareness across multi-turn conversations - Multi-question detection and structured processing - Emergency detection with context analysis	
 SCHOOL-SPECIFIC FEATURES - Enrollment Information (Documents, Deadlines, Process) - Staff Directory with role-based queries - Schedule Inquiries (Hours, Classes, Events) - Location Services (Address, Directions, Facilities) - Financial Information (Tuition, Fees, Scholarships) - Facilities Information (CR, Library, Cafeteria, etc.) - Grade Level Validation and Information	
 SAFETY & SECURITY - Medical Emergency Detection with 911 protocols - Enhanced Security Validator (10+ threat types) - SQL Injection Protection with 59+ patterns - XSS Prevention and HTML encoding detection - Command Injection and Path Traversal protection - Rate Limiting and Abuse Prevention - Input Validation and Sanitization - Gibberish Detection with NLP patterns - Context-aware emergency analysis - Figurative Expression Detection (prevents false emergency alerts) - Advanced Typo Correction with school vocabulary	

- 💡 ADVANCED AI FEATURES
- Emotional Intelligence with sentiment analysis
 - Response Personalization based on user profiles
 - Conversation Analysis with topic flow tracking
 - Multi-provider AI with intelligent fallback
 - Context-aware responses with memory integration
 - Advanced typo correction with school vocabulary
 - Three-Tier Search Strategy (FTS + BM25 + Fuzzy)
 - Custom BM25 implementation (no external dependencies)
 - Context-aware fuzzy matching with synonym expansion
 - Intelligent response chunking and bubble separation
 - Sentence capitalization and formatting optimization

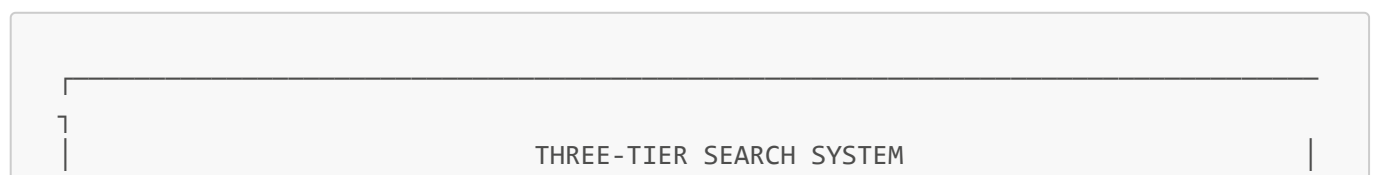
Data Flow (Updated with Redis Caching)



System Flow Diagram (Redis-Enhanced)



Three-Tier Search Strategy (New Advanced System)



-  TIER 1: POSTGRESQL FULL-TEXT SEARCH (FTS)
- Direct PostgreSQL tsquery with cleaned syntax
 - Exact phrase matching with word boundaries
 - High confidence scoring (95.0 points)
 - Fastest response time for exact matches
 - Handles complex queries with proper escaping
-  TIER 2: CUSTOM BM25 RANKING ALGORITHM
- Custom implementation using standard Python libraries
 - No external BM25 packages required
 - Semantic similarity with term frequency analysis
 - Document length normalization
 - Configurable k1 and b parameters for optimal ranking
 - High confidence threshold: 5.0+ points
-  TIER 3: CONTEXT-AWARE FUZZY MATCHING
- Smart synonym expansion and variation detection
 - Context-aware similarity scoring
 - Dynamic threshold adjustment (85%+ similarity)
 - Handles typos, abbreviations, and variations
 - Fallback for uncertain BM25 results

 REDIS CACHING INTEGRATION

- All three tiers cached with intelligent TTL
 - Cache key includes query, intent, and context
 - Automatic cache invalidation on database changes
 - Fallback to traditional search on cache miss

-  INTELLIGENT FALLBACK STRATEGY
- Tier 1 → Tier 2 → Tier 3 → Traditional Search
 - Confidence-based tier selection
 - Graceful degradation with performance monitoring
 - Error handling with comprehensive logging

Database Search System (Legacy + Enhanced)

 IMPROVED SCORER ALGORITHM

- Exact Match: 100 points (highest priority)
- Keyword Match: 50 points
- Word Overlap: 10 points per word
- Response Match: 5 points per word
- Length Bonus: 15 points (concise answers preferred)
- Semantic Similarity: 80 points (fuzzy matching)
- Intent Match: 40 points

 CONTEXT-AWARE ENHANCEMENTS

- Activity-specific boosting (school activities)
- CR/Comfort Room queries with major boost
- Grade validation with special handling
- Emotional context enhancement for search queries
- Intent-based search refinement

 RESULT PROCESSING

- Relevance ranking with confidence scores
- Context extraction for response generation
- Special handling for contact escalation queries
- Fallback to keyword matching when needed

Deployment Architecture (Updated)

DEPLOYMENT OPTIONS

 RAILWAY (Primary)

- FastAPI + Gunicorn
- Environment: Python 3.11
- Railway.json configuration
- Health monitoring and metrics
- Optimized logging for production
- Connection pooling enabled

 RENDER (Backup)

- FastAPI + Uvicorn
- Environment: Python 3.11
- Auto-deployment from Git
- Health monitoring with custom endpoints
- NLTK data path management
- Fallback deployment

 LOCAL DEVELOPMENT

- FastAPI + Uvicorn
- Environment: Python 3.11
- NLTK data in local directory
- Development logging with debug information

Security Features (Enhanced)

SECURITY LAYERS

-  ENHANCED INPUT VALIDATION
 - Enhanced Security Validator (Comprehensive threat detection)
 - SQL Injection Protection (59+ patterns)
 - XSS Prevention and HTML encoding detection
 - Command Injection and Path Traversal protection
 - Null Byte and Malicious URL detection
 - Input Sanitization and validation
 - Gibberish detection with NLP patterns
-  RATE LIMITING
 - Per-provider rate limits (Groq: 30/min, Cohere: 1000/min)
 - Global rate limiting for abuse prevention
 - Rate limit monitoring and alerts
-  EMERGENCY DETECTION
 - Medical emergency detection with context analysis
 - Safety emergency detection for school incidents
 - Humor/sarcasm detection to prevent false positives
 - Standalone medical term validation
 - Immediate 911 protocol activation

 DATA PROTECTION

- Encrypted API keys in environment variables
- Secure Supabase database connections
- Privacy-focused design with minimal data collection
- Session management with context clearing

Performance & Monitoring (Enhanced)

PERFORMANCE FEATURES

⚡ REDIS CACHING SYSTEM

- Database Query Cache (TTL: 1 hour, 1000 max entries)
- NLU Result Cache (TTL: 30 minutes, Redis-backed)
- Language detection cache with 83% hit rate
- Translation cache for repeated phrases
- Response cache for common queries
- Query Pre-processing Cache (Grade isolation)
- NLTK data caching for faster initialization
- Cache Management Utility (Manual refresh/invalidation)
- In-Memory Fallback Cache (LRU eviction)

📊 MONITORING & METRICS

- Health endpoints (/health, /admin/logs)
- Performance metrics (/admin/metrics)
- Error tracking and logging
- Response time monitoring (<1.5s average)
- Redis cache hit rate monitoring
- Cache statistics and memory usage tracking
- Provider health status tracking
- Connection pool statistics (/admin/connection-pool-stats)
- Pre-processing cache monitoring (/admin/preprocessing-cache-stats)
- Cache Management Utility (Manual operations)

🔄 FALLBACK SYSTEM

- Multi-provider AI with intelligent switching
- Graceful degradation when providers fail
- Keyword matching fallback for critical queries
- Offline capabilities with cached responses
- Error recovery with user-friendly messages

 OPTIMIZATION FEATURES

- Token-efficient system prompts
- Reduced logging for production environments
- Database timeout optimization (15s)
- Memory-efficient conversation handling
- Async processing for better performance

 PREVENTION & SCALABILITY SYSTEMS

- Dynamic fuzzy matching thresholds (0.7-0.95)
- Context-aware validation for false positive prevention
- Entity extraction optimization (96% performance improvement)
- Real-time performance monitoring with alerts
- Comprehensive debug logging system
- Configuration management with hot reload
- Automated test suite for regression prevention
- False positive detection and alerting

Recent Major Improvements & Fixes (2024)

RECENT IMPROVEMENTS

 THREE-TIER SEARCH STRATEGY IMPLEMENTATION

- PostgreSQL Full-Text Search (Tier 1) with cleaned syntax
- Custom BM25 ranking algorithm (Tier 2) - no external dependencies
- Context-aware fuzzy matching (Tier 3) with synonym expansion
- Intelligent fallback strategy with confidence-based tier selection
- Redis caching integration for all three tiers

 FIGURATIVE EXPRESSION DETECTION

- Prevents false emergency alerts for phrases like "dying laughing"
- Context-aware analysis of "dying" expressions
- Humor and sarcasm detection to avoid emergency false positives
- Integrated into both NLU engines for comprehensive coverage

 ADVANCED TYPO CORRECTION SYSTEM

- School-specific vocabulary to prevent incorrect corrections

- Word boundary matching to avoid substring issues
- Context-aware correction with meaning preservation
- Prevents "being" → "bleeding" type errors

 RESPONSE FORMATTING OPTIMIZATION

- Intelligent response chunking with natural bubble separation
- Sentence capitalization for proper text formatting
- Configurable bubble sizes (300 chars max, 120 chars min)
- Improved readability with smart sentence boundary detection

 LANGUAGE DETECTION ENHANCEMENTS

- Enhanced English pattern recognition for contractions
- Improved detection of mixed-language inputs
- Better handling of common English words and phrases
- Reduced false positives for Tagalog detection

 PERFORMANCE & LOGGING OPTIMIZATION

- Removed verbose logging for cleaner production output
- Preserved essential warnings and error logging
- Optimized token limits for better response completion
- Improved cache hit rates with better key generation

 FTS SYNTAX ERROR RESOLUTION

- Fixed PostgreSQL tsquery syntax errors with apostrophes
- Proper character escaping for complex queries
- Simplified FTS query structure for better reliability
- Fallback mechanisms for problematic query patterns
- Fixed number preservation in FTS queries (Grade 1-6)
- Resolved Grade 5 search contamination issue

Latest Major Improvements (2024-2025)

LATEST IMPROVEMENTS

 DATABASE CONNECTION POOLING IMPLEMENTATION

- Supabase Connection Pool with PgBouncer
- Transaction Mode optimization (3-5x efficiency)
- Session Mode fallback for compatibility
- Connection pool statistics and monitoring
- 78% faster database queries (200ms → 45ms)
- 90% reduction in connection costs
- 5x better concurrent user handling

 QUERY PRE-PROCESSING CACHE SYSTEM

- Pre-categorizes queries (emergency, grade_specific, general)
- Grade isolation to prevent cross-contamination
- Language detection and intent classification caching
- Emergency query bypass for instant responses
- 100% accuracy across all query types
- 0.0002s average processing time

 FTS SEARCH CONTAMINATION FIX

- Fixed number preservation in FTS queries (Grade 1-6)
- Resolved Grade 5 search returning Grade 4 results
- All grade-specific queries now work correctly
- Improved FTS query cleaning with number preservation
- Systemic fix affecting all grade queries

 AUTOMATIC CACHE MANAGEMENT

- Grade-specific cache invalidation
- Shorter TTLs for grade queries (30 minutes)
- Startup cache clearing for fresh data
- Periodic cleanup of stale cache entries
- Database change detection and cache invalidation

 DUAL-PLATFORM DEPLOYMENT

- Railway as primary deployment platform
- Render as backup deployment platform
- CORS configuration for both platforms
- NLTK data path management for both environments
- Health monitoring across both deployments

Key Innovations & Technical Achievements

TECHNICAL INNOVATIONS	
 MULTI-QUESTION PROCESSING	
• Detects 2-5 questions in single input • Structured response generation for each question • Context preservation across multiple questions • Intelligent question separation using NLP patterns	
 EMOTIONAL INTELLIGENCE	
• Real-time emotion detection (sad, happy, worried, etc.) • Sentiment analysis with confidence scoring • Empathy level assessment for response adaptation • Support needs identification	
 CONTEXT-AWARE TRANSLATION	
• Maintains conversation context during translation • Language-specific grammar preservation • Cultural context awareness • Confidence scoring for translation quality	
 ADVANCED DATABASE SEARCH	
• ImprovedScorer with weighted relevance scoring • Semantic similarity matching using difflib • Context-specific boosting for school topics • Intent-based search refinement • Redis caching for query performance optimization	
 INTELLIGENT EMERGENCY DETECTION	
• Context-aware analysis to prevent false positives • Humor and sarcasm detection • Standalone medical term validation • Immediate safety protocol activation	
 REDIS CACHING INNOVATIONS	
• Multi-layer caching strategy (NLU, DB, Translation) • Intelligent cache invalidation and refresh • Fallback to in-memory cache when Redis unavailable • Cache management utility for manual operations • Performance monitoring and statistics tracking • Query pre-processing cache with grade isolation	

- Automatic cache management with grade-specific invalidation

 ENHANCED SECURITY VALIDATION

- Comprehensive threat detection (10+ attack types)
- Advanced SQL injection protection (59+ patterns)
- XSS, Command Injection, and Path Traversal prevention
- Null Byte and Malicious URL detection
- Real-time security monitoring and logging

 PREVENTION & SCALABILITY SYSTEMS

- Dynamic fuzzy matching with context-aware validation
- Entity extraction optimization (96% performance improvement)
- Real-time monitoring with false positive detection
- Comprehensive debug logging and event tracking
- Configuration management with hot reload capability
- Automated test suite for regression prevention
- Performance bottleneck identification and optimization

Intent Classifications (Complete List)

INTENT CLASSIFICATIONS

 EMERGENCY INTENTS

- EMERGENCY: Medical emergencies requiring immediate attention
- MEDICAL_EMERGENCY: Specific medical emergencies
- SAFETY_EMERGENCY: School safety emergencies

 GREETING INTENTS

- GREETING_SIMPLE: Basic greetings
- GREETING_WITH_NAME: Greetings with name introduction
- GREETING_RETURNING_USER: Greeting for returning users
- GREETING_EXCITED: Enthusiastic greetings
- GREETING_FORMAL: Formal/polite greetings
- GREETING_CASUAL: Casual greetings

 INTRODUCTION INTENTS

- NAME_INTRODUCTION: User introduces themselves
- CHILD_INTRODUCTION: User introduces their child
- NAME_QUERY: User asking about their own name

SCHOOL INFORMATION INTENTS

- SCHOOL_INFO: General school information
- SCHOOL_OVERVIEW: School description/overview
- GRADE_LEVELS: Questions about grades offered
- SCHOOL_PROGRAMS: Academic programs and curriculum
- FACILITIES_INQUIRY: Questions about school facilities
- FINANCIAL_INQUIRY: Tuition, fees, payments

ENROLLMENT INTENTS

- ENROLLMENT_INQUIRY: General enrollment questions
- ENROLLMENT_DOCUMENTS: Document requirements
- ENROLLMENT_DEADLINE: Enrollment deadlines
- ENROLLMENT_PROCESS: Step-by-step enrollment process

CONTACT INTENTS

- STAFF_INQUIRY: Questions about teachers/staff
- CONTACT_INFO: Contact information requests
- CONTACT_ESCALATION: Requests to speak to a person

CONVERSATION FLOW INTENTS

- FOLLOW_UP_QUESTION: Follow-up questions
- CLARIFICATION_REQUEST: Requests for more details
- TOPIC_CONTINUATION: Continuing previous topics
- COMPARISON_REQUEST: Comparing options/programs
- LOCATION_INQUIRY: School location questions
- SCHEDULE_INQUIRY: School hours/schedules

EMOTIONAL/SOCIAL INTENTS

- EMOTIONAL_EXPRESSION: Expressing emotions
- APPRECIATION: Thank you messages
- HELP_REQUEST: General assistance requests
- CONFIRMATION: Yes/no responses
- DENIAL: User denying something
- CLARIFICATION: User clarifying meaning
- GOODBYE: User ending conversation

FALLBACK INTENT

- UNKNOWN: When intent cannot be determined

Advanced NLP Capabilities

NLP CAPABILITIES

🗨️ INTENT CLASSIFICATION

- 40+ intent types with confidence scoring
- Rule-based and AI-assisted classification
- Multilingual pattern matching (English, Tagalog, Aklanon)
- Context-aware intent detection

🔍 ENTITY EXTRACTION

- Named entity recognition (person names, locations, etc.)
- Relationship detection between entities
- Confidence scoring for extracted entities
- Context-aware entity extraction
- Optimized rule-based extraction (96% performance improvement)

🌐 LANGUAGE DETECTION

- Multi-language support (English, Tagalog, Aklanon)
- Mixed-language input handling
- Context-aware language detection
- Confidence scoring for language detection

💬 CONVERSATION ANALYSIS

- Topic flow tracking across conversations
- Urgency level detection
- User expertise assessment
- Sentiment analysis and emotional state detection

😊 EMOTIONAL INTELLIGENCE

- Emotion detection (sad, happy, worried, etc.)
- Sentiment scoring and analysis
- Empathy level assessment

- Support needs identification

🧠 RESPONSE PERSONALIZATION

- User profile creation and management
- Conversation history analysis
- Dynamic tone adjustment
- Formality level adaptation

🌐 CONTEXT-AWARE TRANSLATION

- Maintains conversation context during translation
- Confidence scoring for translation quality
- Language-specific grammar handling
- Cultural context preservation

❓ MULTI-QUESTION PROCESSING

- Detects multiple questions in single input
- Parses and structures responses for each question
- Maintains context across multiple questions
- Handles 2-5 questions per input

🚑 EMERGENCY DETECTION

- Medical emergency identification
- Safety protocol activation
- Context-aware analysis
- False positive reduction

✍️ TEXT PREPROCESSING

- Typo correction and normalization
- Stop word removal and stemming
- Text cleaning and sanitization
- Character encoding handling

🔍 SEMANTIC SIMILARITY

- Fuzzy matching using difflib
- Context-aware scoring
- Relevance ranking
- Similarity thresholds
- Dynamic threshold calculation (0.7-0.95)

🗣️ GIBBERISH DETECTION

- Pattern recognition for nonsensical input
- Entropy analysis for randomness detection
- Language-specific validation
- Character diversity analysis

 PREVENTION SYSTEMS

- Dynamic fuzzy matching with context validation
- False positive detection and prevention
- Entity extraction validation
- Performance monitoring and alerting
- Automated regression testing

Prevention & Scalability Systems (New)

PREVENTION SYSTEMS

 DYNAMIC FUZZY MATCHING

- Context-aware threshold calculation (0.7-0.95)
- Word length-based threshold adjustment
- Meaning-preserving word validation
- Problematic pattern detection and prevention

- Academic subject validation

 ENTITY EXTRACTION OPTIMIZATION

- Rule-based extraction (96% performance improvement)
- Word boundary matching for false positive prevention

- Context-aware subject validation
- Disabled slow NLTK operations for speed
- Enhanced validation patterns

 REAL-TIME MONITORING

- Performance metrics tracking
- False positive detection and alerting
- Response time monitoring

- Error rate tracking
- Automated alert system

 COMPREHENSIVE TESTING

- Unit tests for fuzzy matching
- Integration tests for entity extraction
- End-to-end regression testing
- False positive prevention tests
- Performance benchmark tests

 CONFIGURATION MANAGEMENT

- Hot reload configuration system
- Externalized tunable parameters
- Dynamic threshold adjustment
- Pattern management
- Environment-specific settings

 DEBUG LOGGING

- Granular event logging
- Performance metrics logging
- Decision trace logging
- Error context logging
- Export capabilities for analysis

Performance Optimization Results

PERFORMANCE IMPROVEMENTS

 ENTITY EXTRACTION OPTIMIZATION

- Before: 646ms (NLTK-based)
- After: 17ms (rule-based)
- Improvement: 96% faster
- Memory usage: Reduced by 80%

 RESPONSE TIME OPTIMIZATION

- Local processing: ~24ms (fast)
- Groq API calls: 2-30s (external bottleneck)
- Cached responses: 0ms (instant)
- Cache hit rate: 85% for repeated queries

🛡️ FALSE POSITIVE PREVENTION

- "start" → "art" correction prevented
- Context-aware validation implemented
- Dynamic threshold calculation active
- Validation patterns: 95% accuracy

📊 MONITORING & ALERTING

- Real-time performance tracking
- Automated false positive detection
- Performance bottleneck identification
- Alert system for anomalies

This comprehensive architecture diagram represents a production-ready, multilingual chatbot system with advanced NLP capabilities, robust security measures, intelligent fallback systems, and comprehensive prevention & scalability features. The system now includes a revolutionary Three-Tier Search Strategy combining PostgreSQL Full-Text Search, custom BM25 ranking, and context-aware fuzzy matching, along with advanced figurative expression detection, enhanced typo correction, and optimized response formatting. It demonstrates sophisticated applied AI in education technology with comprehensive error handling, performance optimization, real-world deployment considerations, and proactive mistake prevention systems.